Julius Zhang

juliuszh@alumni.stanford.edu

I am interested in identifying structures and developing new primitives for practical problems.

Experience

a16z — Researcher, 2025.4 - Present

- Building Jolt, open-source zero-knowledge virtual machine in Rust.
- Implemented and benchmarked torus compression; proved optimality with a conceptual reframing via Hilbert 90.
- Audited elliptic-curve optimizations, formalizing folklore results into rigorous proofs.
- Performance engineering across the proving stack, including sum-check optimization.

Millennium — Researcher, 2024.6 - 2025.4

- Convex optimization and robust statistical estimation for real-time trading systems.

Stellar Development Foundation — Consultant, 2023.6 - 2023.9

- Byzantine fault tolerance, C++, stellar-core.
- Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.

Distributed Systems and TCS Research, Stanford University — Researcher, 2023.9 - 2024.6

- Under David Mazières, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules.
- Research PCP under Aviad Rubinstein, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.

Topology Research, Stanford University — Researcher, 2020.7 - 2022.5

- Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections.
- Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.

Selected Papers and Manuscripts

- Julius Zhang. *The Trace-Zero Subgroup and the Relative Trace for BN254*, 2026. [PDF]
- Julius Zhang. *Complex Cobordism and Formal Group Law*, Manuscript (Undergraduate Honours Thesis), Stanford University, 2024. [PDF]

Talks

- **p-Ordinary Cohomology for** $\mathrm{SL}(2)$ **over Number Fields (after Hida)**,
  Hida Theory Seminar, Columbia University, 2026.

Education

Courant Institute of Mathematical Sciences, New York University, computer science *PhD.* — starting 2026.9

- Verifiable systems and theoretical computer science.

Stanford University, computer science *M.S.* and mathematics *B.A.* — 2019.9 - 2024.6

- General Yaowu Wang Scholarship.

Skills

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, homotopy theory, Floer theory, gauge theory, concentration inequalities.